



Attention Mechanisms

Chapter 3: The Core of the Transformer Architecture

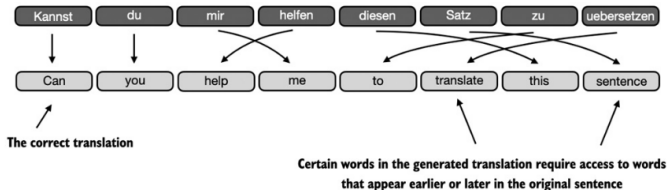
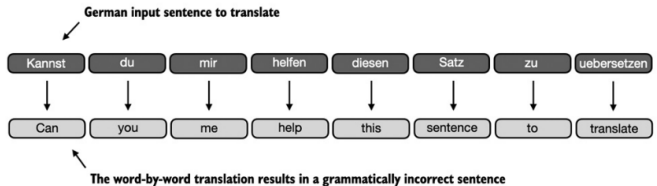
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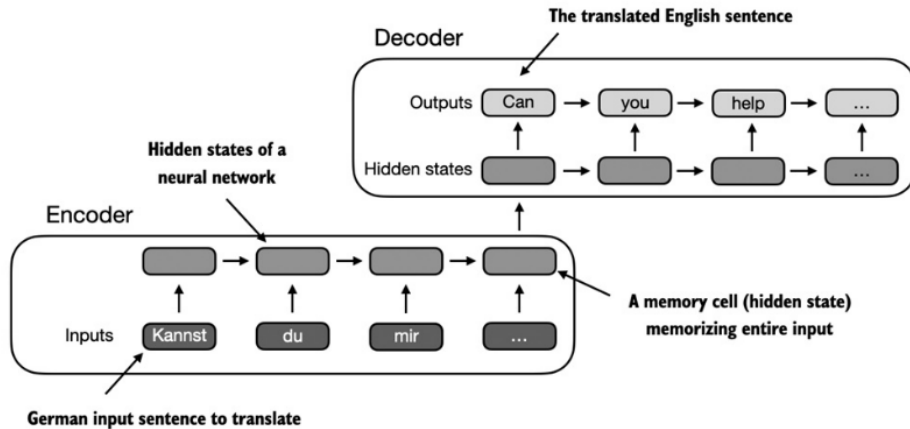
The Motivation for Attention

- The task



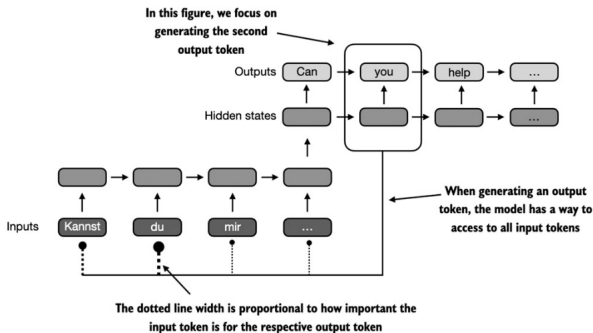
The Motivation for Attention

- **Problem with RNNs:** Difficulty in capturing long-range



The Motivation for Attention

- **The Attention Concept:** Allows the model to focus on different parts of the input sequence regardless of their distance.
- **Key Insight:** Instead of a fixed-size hidden state, we compute a weighted sum of all input tokens' representations.



The paper that made the revolution

arXiv:1706.03762v7 [cs.CL] 2 Aug 2023

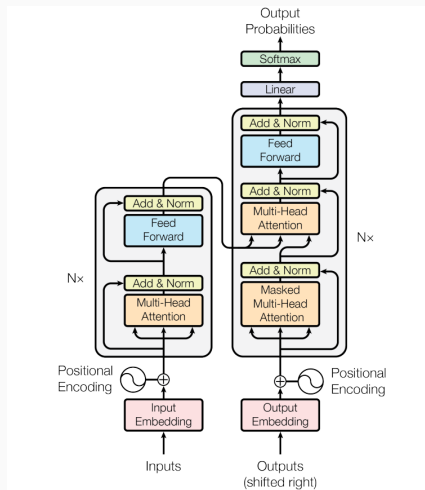
Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

The transformer architecture



Attention mechanism (Vaswani et al. 2017)

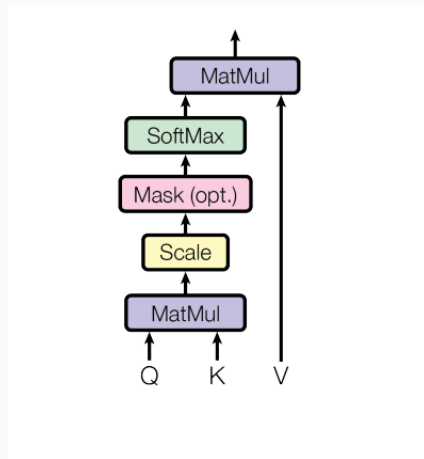


Figure 1: Scaled dot-product attention

Scaled Dot-Product Attention: The Math

The core operation involves three learnable matrices: **Query (Q)**, **Key (K)**, and **Value (V)**.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (1)$$

- **Dot-Product (QK^T)**: Measures similarity between queries and keys.
- **Scaling ($1/\sqrt{d_k}$)**: Prevents the softmax from entering regions with extremely small gradients when d_k is large.
- **Softmax**: Normalizes scores into probabilities (weights summing to 1).

Self-Attention vs. Cross-Attention

Self-Attention:

- Q, K, V all originate from the same input sequence.
- Used in Encoders and Decoders to relate different positions of the same sequence.

Cross-Attention:

- Q comes from one sequence (e.g., decoder), while K and V come from another (e.g., encoder output).
- Essential for tasks like Machine Translation.

Causal Attention: Masking the Future

In GPT-like models (Decoder-only), a token should only attend to previous tokens.

- **Implementation:** Apply a triangular mask (lower triangular matrix) to the scores before the softmax.
- **Value:** Future positions are set to $-\infty$, resulting in 0 probability after softmax.

Multi-Head Attention (MHA)

Why use multiple heads?

- Allows the model to jointly attend to information from different representation subspaces at different positions.
- **Process:**
 1. Split Q, K, V into h heads.
 2. Perform attention in parallel.
 3. Concatenate the results and apply a linear projection.

[Image of multi-head attention architecture]

Computational Efficiency and Complexity

- **Self-Attention Complexity:** $O(L^2 \cdot d)$, where L is sequence length and d is embedding dimension.
- **Comparison:**
 - RNNs are $O(L \cdot d^2)$ but sequential.
 - Transformers are parallelizable but quadratic with respect to length.
- This quadratic growth is why extending the **Context Window** is a major research challenge in LLMs.

From Math to Code: PyTorch Structure

```
class MultiHeadAttention(nn.Module):
    def __init__(self, d_in, d_out, context_len,
                  dropout, num_heads):
        super().__init__()
        self.W_query = nn.Linear(d_in, d_out, bias=
                                   False)
        self.W_key = nn.Linear(d_in, d_out, bias=False
                                )
        self.W_value = nn.Linear(d_in, d_out, bias=
                                   False)
        # ... rest of the logic
```

Crucial: Biases are often omitted in Q, K, V linear layers in modern implementations to simplify the geometry of the projection.

Next Chapter: Implementing a GPT Model from Scratch